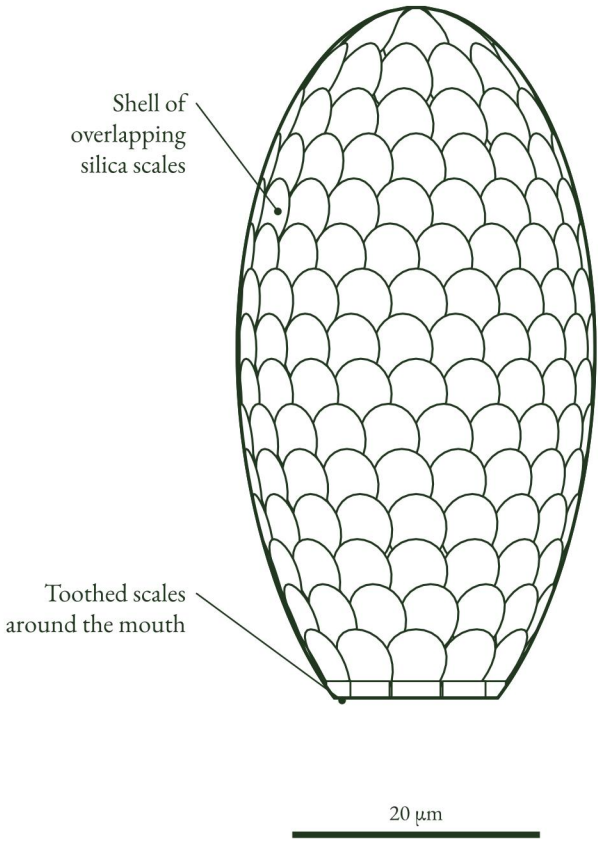
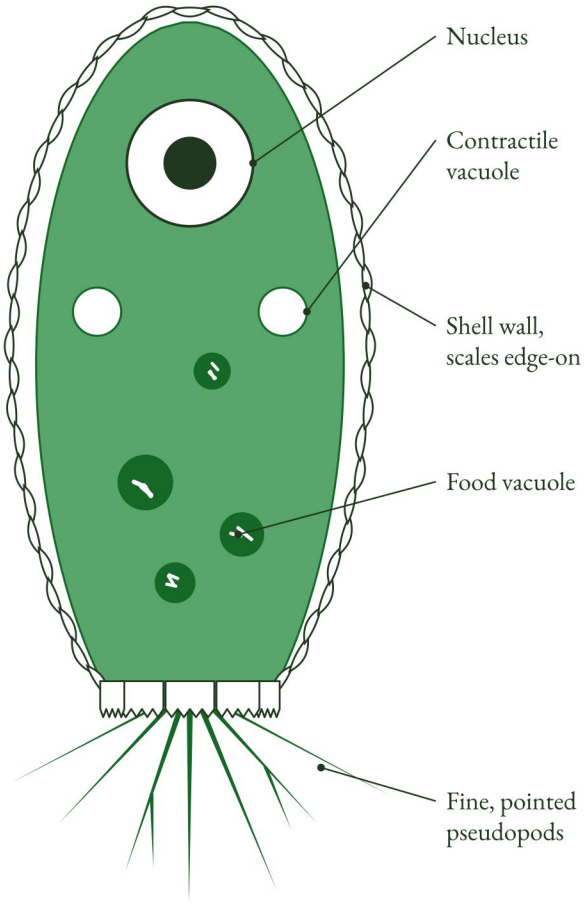
Euglypha

A shelled amoeba that builds its house from glass

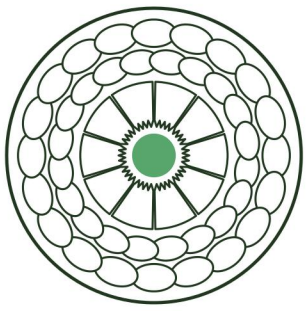
Testate amoeba · Rhizaria · Cercozoa · Euglyphida



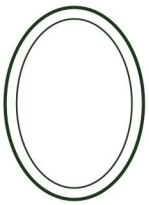
A Shell surface



B Living cell, mid focus



C The mouth, end-on



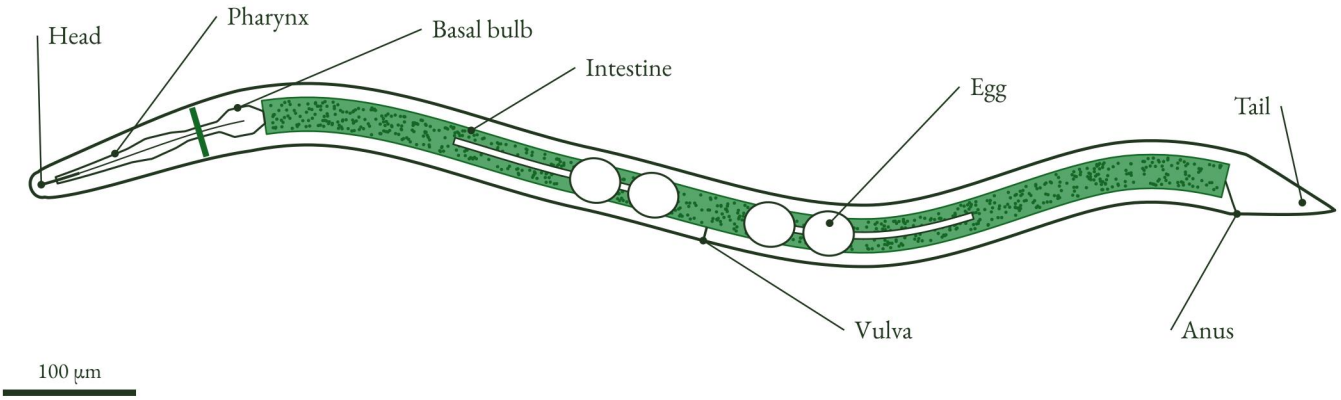
D One scale

Size	Shell about 30 to 100 µm long, depending on the species.
Where	Moist soil, leaf litter and moss.
Eats	Bacteria, fungi and small algae, caught with fine pointed pseudopods.
Shell	Silica scales the cell makes itself. It stores a full new set before it divides.
At 400x	A clear, glassy shell. Focus on the surface for the scale pattern, then on the middle for the cell. Look for the toothed mouth.

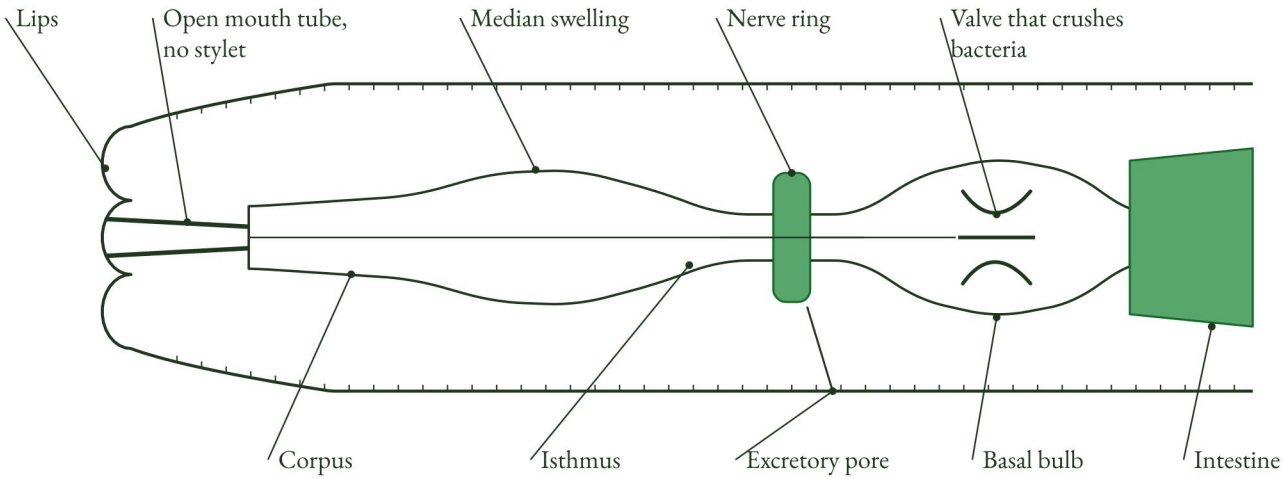
Bacterial-feeding nematode

A roundworm that grazes on bacteria and releases nitrogen for plants

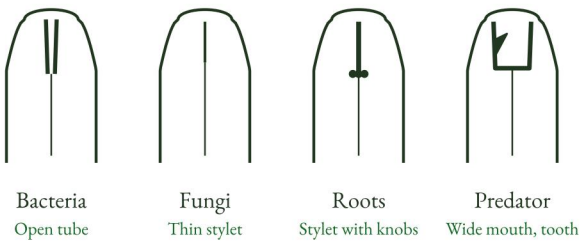
Nematoda · Rhabditida · for example Rhabditis, Cephalobus, Acrobeles



A Whole worm, female, about 1 mm



B Head and pharynx, where you identify it



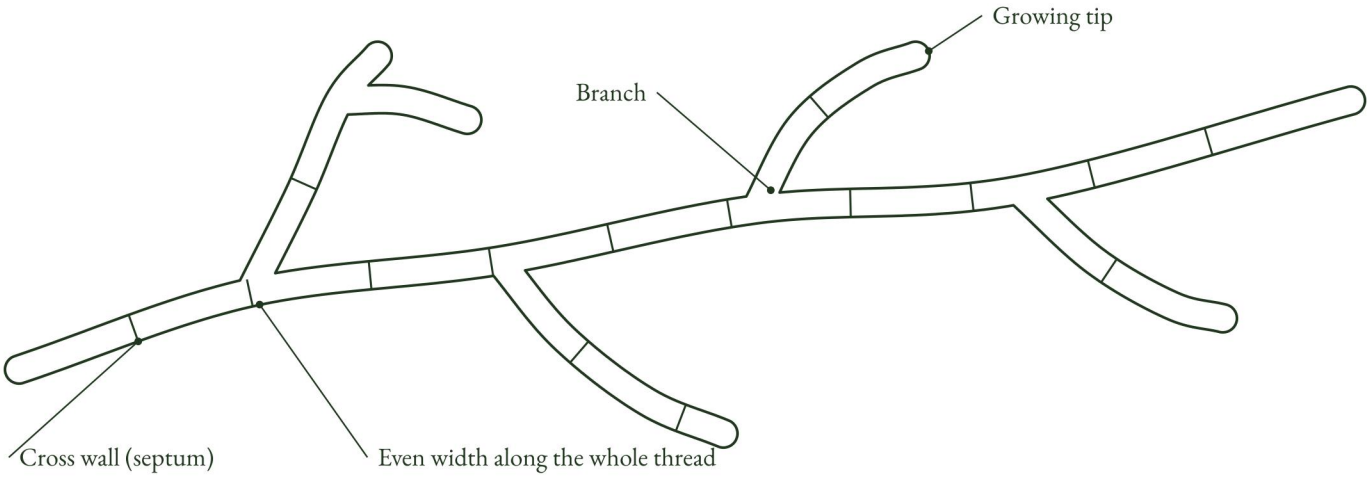
C What each group eats, read from the head

Size	Most are 0.3 to 2 mm long.
Where	Water films around soil particles, around roots and in compost.
Eats	Bacteria, sucked in through the open mouth and crushed in the basal bulb.
Role	It eats more nitrogen than it needs and releases the rest as ammonium, a form plant roots can take up.
At 400x	Look at the head: an open tube, no stylet, and a round bulb that pumps as it feeds.

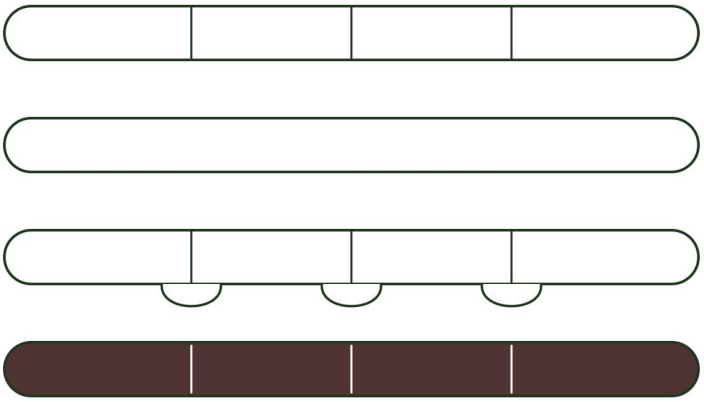
Fungal hyphae

The threads that hold soil together

Fungi · hyphae are the feeding threads, a network of them is mycelium



A A septate hypha, branching, about 400x



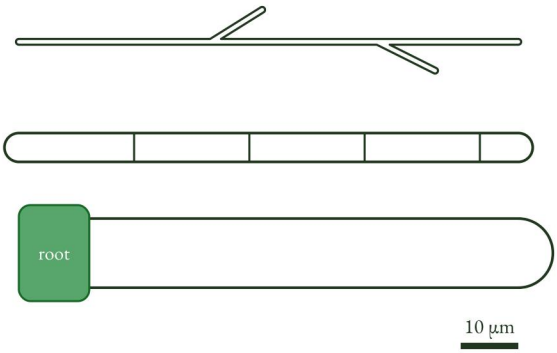
Septate
Cross walls at intervals. Most soil fungi.

Aseptate
No cross walls. Arbuscular mycorrhizal fungi, Mucor.

Clamp connection
A hook over each cross wall. Basidiomycetes.

Brown (pigmented)
Dark wall, often older or decay-resistant hyphae.

B Four kinds to tell apart



Actinobacteria
About 1 µm. Much thinner, branches at sharp angles.

Fungal hypha
About 2 to 10 µm. Even width, cross walls or none.

Root hair
About 10 µm or wider. Grows out of a root, no branches.

C Do not confuse it with, drawn at the same scale

Eats	Dead plant material, even tough lignin and cellulose. Mycorrhizal fungi get sugars from roots.
Role	Binds soil crumbs together, holds nutrients in the soil, and reaches water and nutrients for roots.
At 400x	Look for an even-width tube with cross walls or none. Measure the width against a bacterium.